TAXONOMY OF WORK · SECTION B · SKILL TAXONOMY

Sprint 2 Extraction Plan

Turn each USAJOBS posting into a list of atomic skills, then prove how well that works on 100 postings we label by hand.

Team of 3 | 4 days | Due ≈ Thu 9 Oct 2026 | Dataset: abigailhaddad/usajobs-scraping

What we hand in

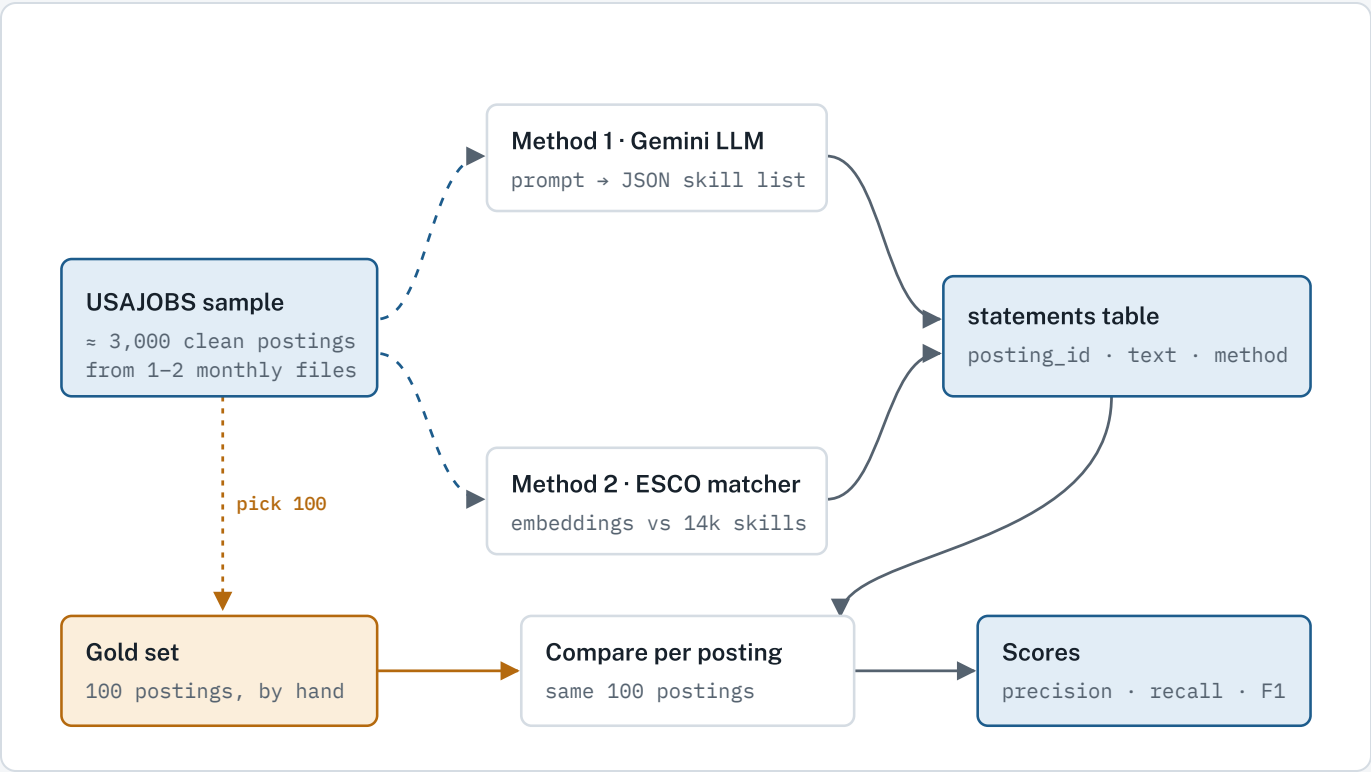
01 · statements table

One row per skill, linked to its posting, with `kind = 'skill'`. Lives in the shared database.

02 · quality report

Method and why, precision / recall / F1 on 100 gold postings, prompts used, LLM token counts.

The whole sprint in one picture



We run extraction on every posting in the sample. Only the 100 gold postings are used to score it. No screenshots: the dataset already gives us the text.

What "extraction" means

POSTING TEXT (QUALIFICATIONSUMMARY)

Specialized experience includes: using SQL and Python to analyze program data; building dashboards in Tableau or Power BI; applying statistical methods; and presenting findings with strong oral communication. Bachelor's degree preferred.

OUTPUT · 6 ATOMIC STATEMENTS

- SQL
- Python
- Tableau
- Power BI
- statistical methods
- oral communication

One skill per row. "Build dashboards" is a task (Section A). "Bachelor's degree" is a qualification, not a skill. We write these rules down so all three of us label the same way.

The dataset

3.2M
federal postings, Sep 2013 – Sep 2026

10.7 GB
parquet, one file per month. We download 1–2 files only.

0
ready-made skill columns. That is why Sprint 2 exists.

FIELD	WHAT IS IN IT	OUR USE
`qualificationSummary`	Specialized experience, required knowledge	Main skill source
`requirements`	Conditions + qualifications	Skill source
`howYouWillBeEvaluated`	Often lists competencies, e.g. "Oral Communication", "Problem Solving"	Near-ready skills
`majorDuties`	What the person will do	Section A (tasks)
`usajobsControlNumber`	Unique id per posting	Our `posting_id`

> **Watch out:** some postings are unfilled HR templates ("[Insert Duties]") or only link to another site. We filter these out before extraction.

Two methods, compared

The course asks us to justify our method. Running two and comparing them is the cleanest justification.

METHOD 1 · LLM

Gemini free tier (Google AI Studio)

Send the skill fields with a few examples. Ask for a JSON list of atomic skills.

- Finds skills phrased in any way
- Recommended default in the course brief
- Rate limits: needs caching and retry
- Can invent skills that are not in the text

METHOD 2 · EMBEDDING MATCH

ESCO skill matcher (Zhang et al. 2022)

Embed each phrase and each of ESCO's ~14k skills. Keep matches above a similarity threshold.

- Free, offline, no quota
- Output already uses ESCO names, which helps Sprint 4
- Misses skills that ESCO lacks
- Threshold needs tuning

How we get a score

```
Posting #1042
  Gold (us):   SQL, Python, Tableau, statistical methods
  Extractor:   SQL, Python, Excel

  ✓ SQL  ✓ Python   ✗ Excel (extra)   – Tableau, statistical methods (missed)
```

PRECISION

2 / 3 = 67%

How much of what we extracted is right.

RECALL

2 / 4 = 50%

How much of the true list we found.

F1

57%

One number that balances both. Average over all 100 postings.

Expect modest numbers. In the Myronenko 2024 paper, GPT-4 reached only about 46% F1 on skills with exact matching. We report both exact and partial (fuzzy) matches.

Four-day plan

A Data B LLM C ESCO + scoring ★ Everyone

DAY 1

Data + rules

- A Download 1–2 months, drop junk, load ≈3,000 postings into SQLite
- B Get a Gemini API key, write the first prompt, test on 5 postings
- C Download ESCO skills, build the labeling sheet
- ★ Agree on labeling rules (what counts as a skill)

DAY 2

Gold set

- ★ Each person labels ≈34 postings (100 total)
- B Add caching + rate-limit retry, run Gemini on the sample
- A Draft the statements schema with Section A

DAY 3

Method 2 + scoring

- C Build the ESCO matcher, tune the threshold
- C Write the scoring script (precision, recall, F1)
- A Write both outputs into the statements table
- B Log tokens, save prompts for the report

DAY 4

Report

- ★ Compare scores, pick the winner (or combine both)
- ★ Write the quality report with errors and examples
- ★ Final check, submit

Roles A, B and C are placeholders. Pick who takes which on our first call.

Papers we found

METHOD 2

Skill Extraction from Job Postings using Weak Supervision

Zhang, Jensen, van der Goot, Plank (2022). Uses ESCO skills plus embeddings to find skills without hand-labeled training data.

SCORING

Enhancing Skill Extraction Using Synthetic Data and Advanced LLMs

Myronenko, Pylieva (UCU, 2024). GPT-4 reached about 46% F1, and fine-tuned BERT about 52%. It sets realistic expectations, and it warns that exact-match scoring is strict.

METHOD 1

[Comparative Analysis of LLMs in Structured Information Extraction from Job Postings]

Springer, 2024. Compares commercial and open LLMs on extracting hard and soft skills. It supports comparing Gemini with a Groq model if we have time.

Ground rules

Cache every LLM call to disk. A re-run costs zero tokens.

No API keys in Git. Keep them in environment variables.

Log token counts. They go in the sprint report.

Public data only. No LinkedIn or Glassdoor.

Open questions for our first call

Q1 Who takes role A, B and C?

Q2 Do we label the gold set from scratch, or correct the LLM output? (Correcting is faster, but we must say so in the report.)

Q3 Which months do we sample? A recent one (2026) is the best match for current skills.

Q4 Does Section A agree on the statements table columns?